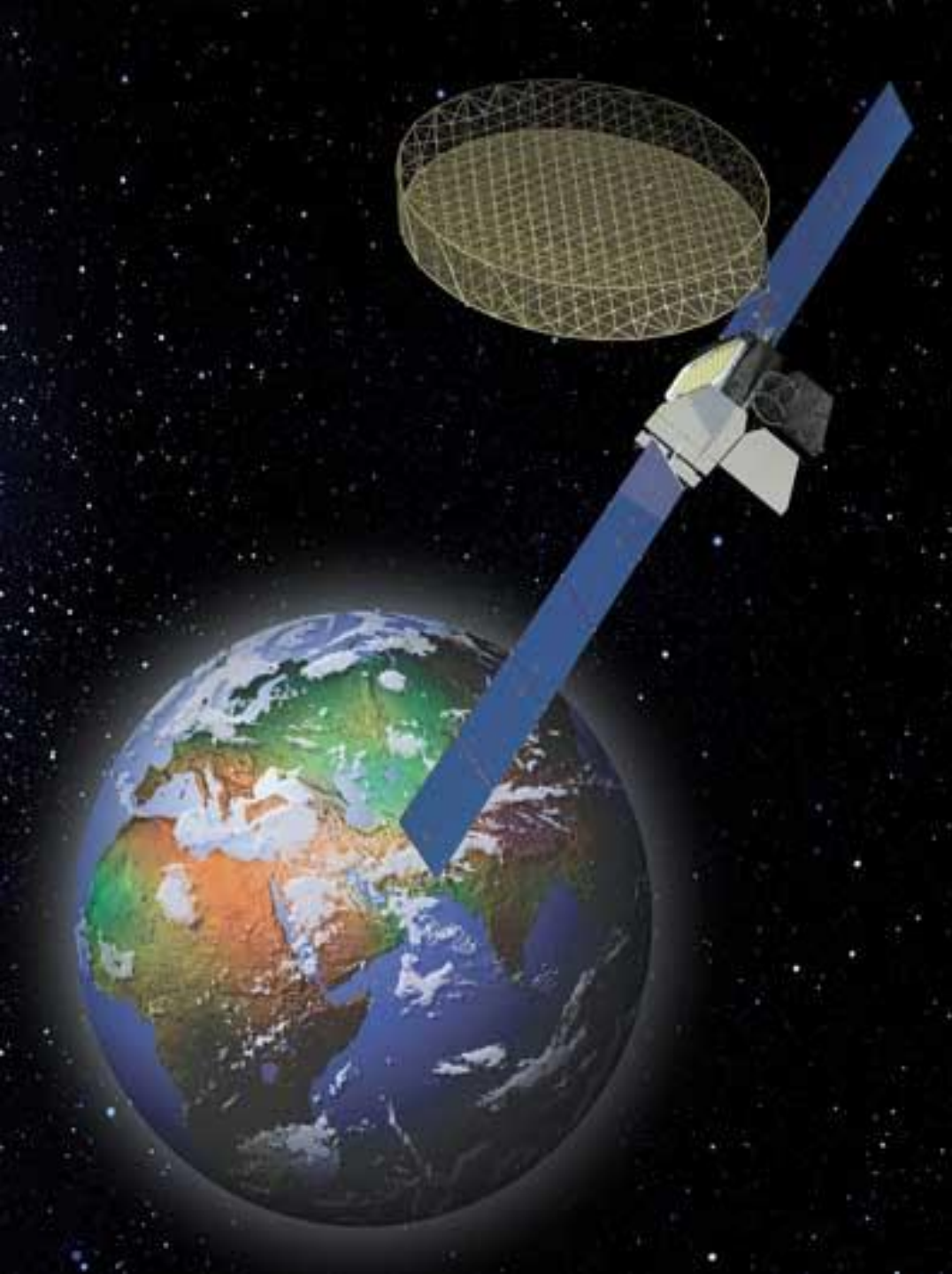




Thuraya 2

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Technical Overview of

ECSE-4500 Satellite Communications · Spring 2026

Thuraya-2

Geosynchronous Mobile Satellite System

35,786 km

orbital altitude

44° East

GEO slot

128

feed elements

200+

spot beams

2.3 B

people served

Presentation Outline

- 01** Introduction - *What is Thuraya-2 · Why it exists · How it works*
- 02** Orbital Configuration - *GEO slot · Coverage footprint · Propagation delay*
- 03** Frequency Allocation - *L-band user links · C-band feeder · Why L-band*
- 04** System Architecture - *Space, ground & subscriber segments · Tx/Rx signal flow*
- 05** Satellite Subsystems - *Power · Payload · AOCS · TTC&M · Thermal*
- 06** Link Budget – *Calculations · Analysis*
- 07** Spot Beams - *Function and purpose*
- 08** Future and Conclusion - *Thuraya 4-NGS · Space42*

The Connectivity Gap

Most of Earth's surface

has no terrestrial cellular coverage — oceans, deserts, mountains, and rural regions are dark zones

2.3 billion people

across Europe, Middle East, Africa, and South Asia need affordable mobile communication

Field operations

Aid workers, maritime crews, military personnel, and remote workers need reliable voice wherever they are

Thuraya-2 solves this with a single geosynchronous satellite — direct voice and data to a 2-watt pocket handset, no infrastructure required.

Thuraya-2 at a Glance

Launch & History

- 1997** Thuraya Telecom founded in UAE as satellite operator
- 2003** Launched June 10 aboard Zenit-3SL from Sea Launch equatorial platform
- 2003** Boeing BSS-702 bus · 12-year design life · 44° East GEO slot
- 2007** Peak subscriber base exceeding 240,000 users, ~\$200M annual revenue
- 2024** Thuraya-3 anomaly → Thuraya-2 reconfigured to cover Southeast Asia
- 2026** Still operational — 22+ years, double its design life

How a Voice Call Works

1

Handset transmits

L-band uplink at 1.6265–1.6605 GHz, 2 W max power

2

Satellite receives

128-element phased array at 35,786 km altitude

3

Onboard routing

DSP routes call — to another handset or to gateway

4

C-band feeder

Signal relayed to Sharjah, UAE gateway on C-band

5

PSTN connection

Gateway connects call to public telephone network

Total one-way propagation delay: ~120 ms · Round-trip: ~240 ms · Mitigated by echo cancellation at Sharjah gateway

Orbital Configuration & Coverage

Orbit type	Geosynchronous (GEO)
Altitude	35,786 km
Orbital slot	44° East longitude
Inclination	6.3° (inclined GEO)
Orbital speed	3.07 km/s
Design life	12 yr (operating 22+ yr)

No inter-satellite handoff required — satellite is fixed relative to Earth

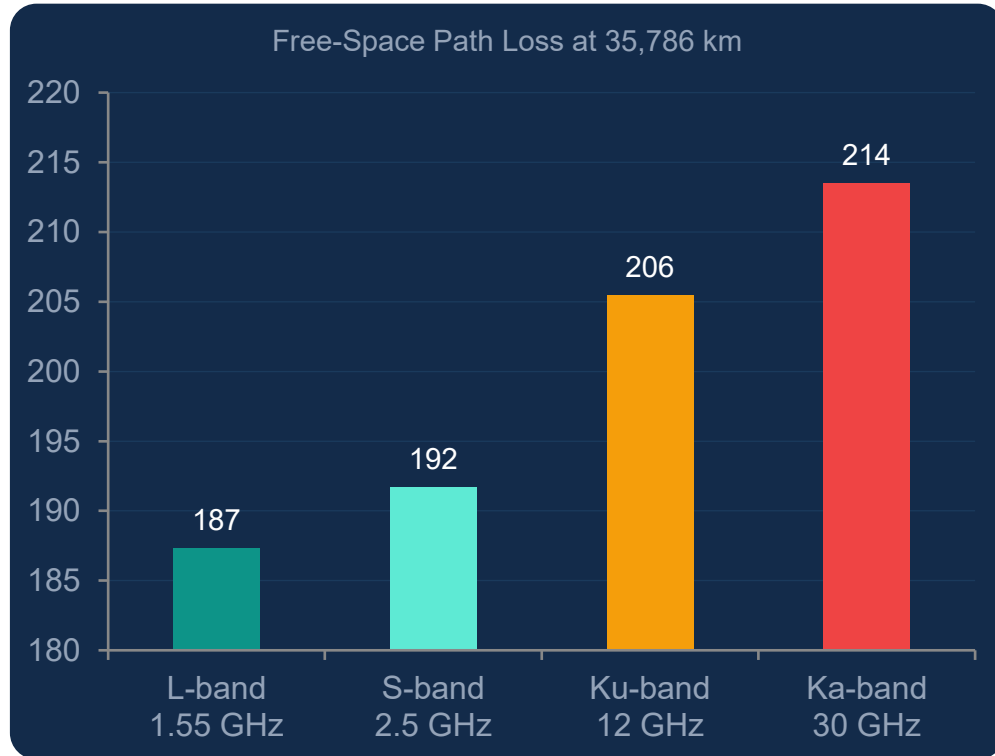
Coverage footprint

Primary: Europe · Middle East · Africa · South Asia · CIS
Post-2024: Southeast Asia added (Thuraya-3 anomaly)
~2.3 billion people in primary footprint



Fig. Thuraya 2 coverage

Frequency Allocation



L-band chosen for Thuraya-2 user links

L-Band — User Links

Downlink: 1.525 – 1.559 GHz

Uplink: 1.6265 – 1.6605 GHz

Bandwidth: 34 MHz · Rain fade: <0.1 dB

18 dB better path loss than Ku-band

C-Band — Gateway Feeder Links

1.27 m round C-band antenna

2 active + 2 spare elements

Links satellite ↔ Sharjah gateway

2 W handset at Ku-band would need 63× more transmit power — impossible for a pocket device

System Architecture — Three Segments

Space Segment

- Boeing BSS-702 bus
- Two solar wings · 11 kW at EOL
- AstroMesh reflector: 12.25×16 m
- 128-element L-band feed array
- 128 × 17 W SSPAs (2,176 W RF total)
- C-band antenna for gateway links
- Onboard digital signal processor

Ground Segment

- Sharjah, UAE ground gateway
- Network Operations Center (NOC)
- Satellite control facility
- C-band feeder link to satellite
- PSTN / internet connection
- Traffic routing & management
- Stationkeeping command uplinks

Subscriber Segment

- TDMA / GMR-1 air interface
- Standard GSM SIM compatible
- 2 W max transmit power
- Built-in GPS receiver
- Beam steering assistance to satellite
- ThurayaIP: up to 444 kbps
- Voice: AMBE codec at 4.8 kbps

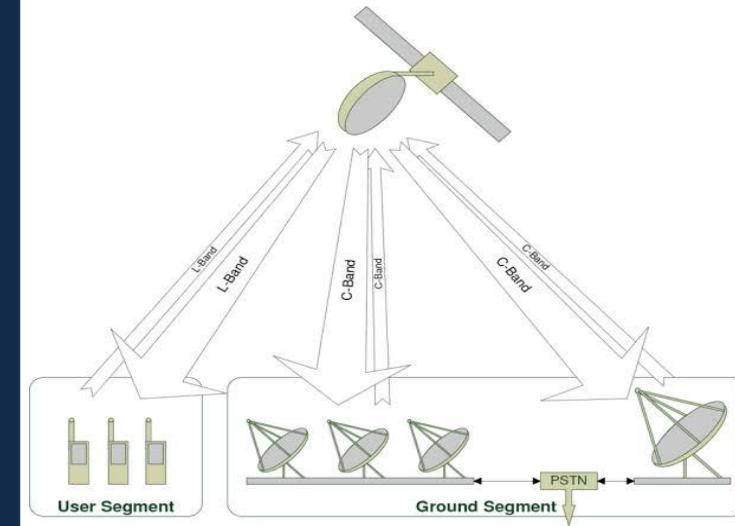


Fig. Thuraya 2 three segment system arch.

Handset → L-band uplink (1.6265 GHz, 2 W) → Satellite → C-band feeder → Sharjah Gateway → PSTN

Satellite Subsystems

Power System

11 kW · 328 A-hr

Two solar wings with 5 cells each generate 11 kW at EOL. Battery pack (328 A-hr) powers the satellite through eclipse periods up to 72 min/day.

Communication Payload

128 elem · 2,176 W RF

12.25×16 m AstroMesh reflector + 128-element feed array. 17 W SSPAs. Digital beamforming creates 200+ steerable spot beams. Supports 13,750 simultaneous voice circuits.

AOCS

3-axis stabilized

Body-stabilized BSS-702 with momentum wheels. Bipropellant stationkeeping thrusters. Operating 22 yr vs 12 yr design life — conservative fuel loading.

TTC&M

S-band · Sharjah

S-band command/telemetry separate from payload. Ranging for orbit determination. Used to reconfigure coverage in 2024.

Thermal Control

Passive + active

N/S radiator panels + loop heat pipes from SSPAs. MLI blankets. Heaters for battery/propellant during eclipse.

Link Budget – Uplink and Downlink

Uplink

Parameter	Value	Notes
TX Power	+3 dBW	From user's handheld device
TX Antenna Gain	+0 dB	Handset antenna
RX Antenna Gain	+44.6 dB	AstroMesh antenna on Thuraya-2
Path Loss	-187.7 dB	1.6265 GHz at GEO distance
Other Losses	-3.5 dB	Atmospheric absorption, polarization mismatch, etc.
RX Power	-143.6 dBW	Sum of other terms
C/N up	+14.4 dB	25.6 dBK system noise temp, 31.5 kHz BW

Downlink

Parameter	Value	Notes
TX Power	+33.4 dBW	Max possible from Thuraya-2's SSPAs
TX Antenna Gain	+44.1 dB	AstroMesh antenna on Thuraya-2
RX Antenna Gain	+0 dB	Handset antenna
Path Loss	-187.3 dB	1.55 GHz at GEO distance
Other Losses	-3.5 dB	Atmospheric absorption, polarization mismatch, etc.
RX Power	-112.8 dBW	Sum of other terms
C/N down	+43.0 dB	27.8 dBK system noise temp, 31.5 kHz BW

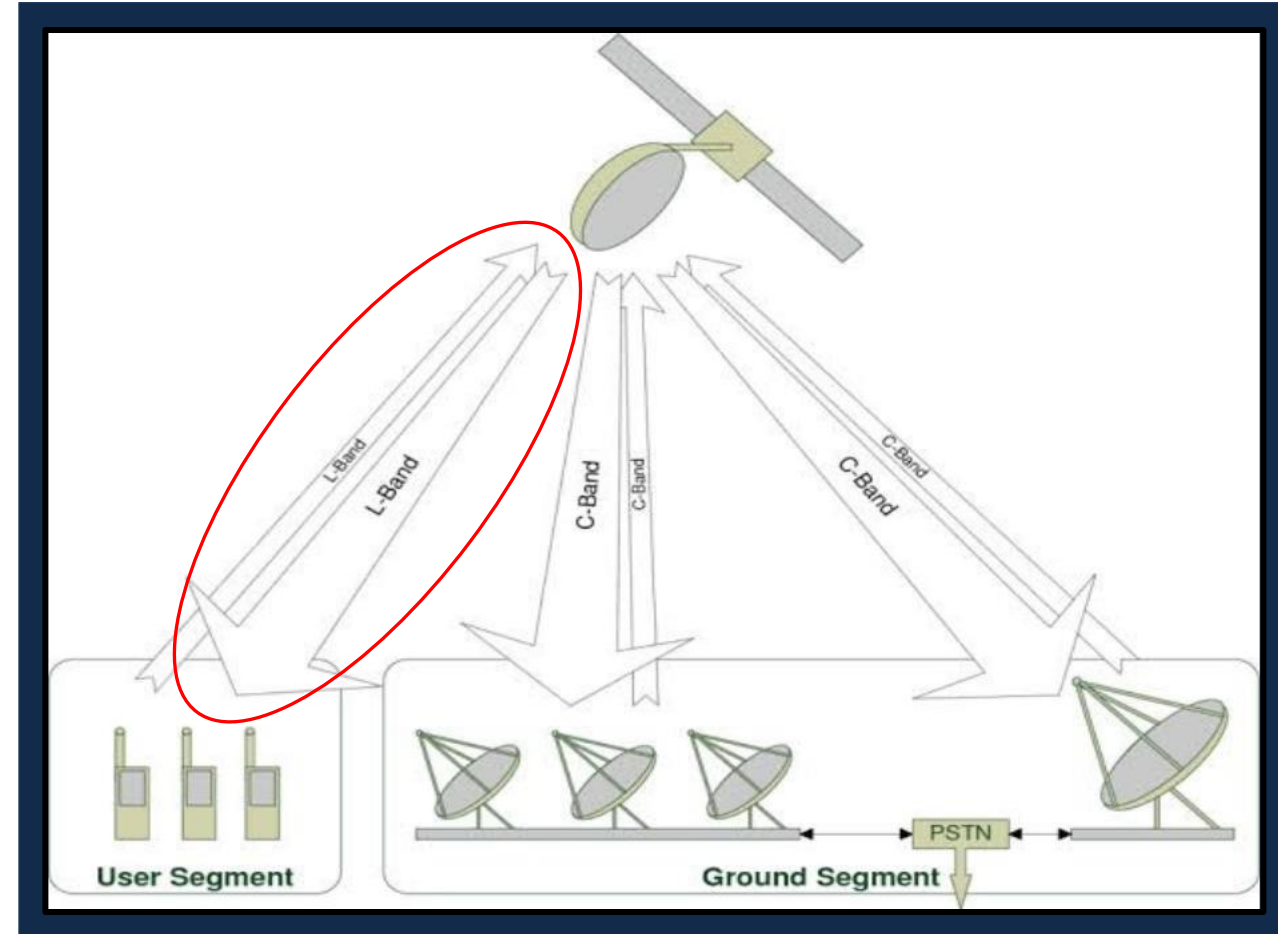


Fig. Thuraya-2 System Architecture

Link Budget – Transmit Power

Thuraya-2 Satellite

- AstroMesh reflector
- Over 2 kW total transmit power

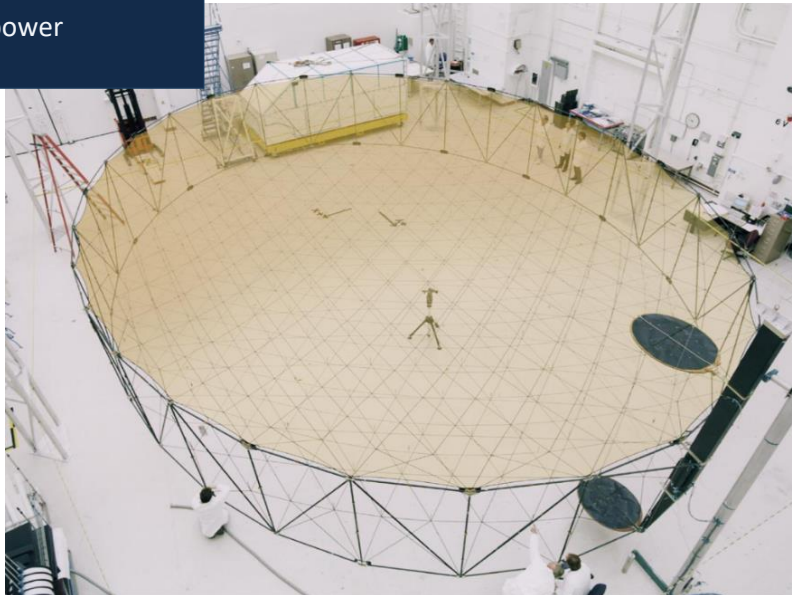


Fig. An example of an AstroMesh reflector

Handheld Satellite Phone

- Omnidirectional antenna
- Only 2 W transmit power



Fig. Thuraya XT-DUAL Satellite Phone

Differences in transmit power account for the largest difference in uplink vs downlink. This allows the transmit power from the satellite to be divided between user links, primarily using spot beams.

Spot Beams



Fig. Map of Thuraya-2 spot beam ID

Satellite-based cellular coverage

- Up to 351 narrow, steerable spot beams are formed by onboard digital beamforming.
- Each beam serves a specific geographic region instead of illuminating the full footprint.

Why it matters

- Concentrates RF power and bandwidth where users are located.
- Dynamic beams can be adjusted on orbit as traffic demand changes.
- Enables up to 13,750 simultaneous voice circuits across the service region.

Key idea: spot beams let Thuraya-2 reuse spectrum and focus capacity over high-demand regions, making a GEO satellite behave like a cellular system.

Future Developments — Extended Lifespan



Fig. Thuraya-2 communication satellite

Why Thuraya-2 remained important

- Designed for about 12–15 years, Thuraya-2 is still operational more than two decades after launch.
- After the Thuraya-3 payload anomaly in April 2024, Thuraya-2 was reconfigured to restore service in affected regions.
- Coverage support was extended toward Southeast Asia and parts of China for critical voice and packet services.
- This demonstrates the value of steerable spot beams and onboard payload reconfiguration even late in mission life.

Takeaway: longevity + flexibility turned Thuraya-2 into an operational backup platform when Thuraya-3 failed.

Future Developments — What Comes Next



Fig. Thuraya 4-NGS next-generation satellite

4.2 Thuraya 4-NGS

- Launched on Jan. 3, 2025 aboard Falcon 9 and positioned at 44°E to replace the aging Thuraya-2.

4.3 Service Migration

- Voice and packet services were migrated during a Nov. 18–19, 2025 maintenance window with only a brief interruption.

4.4 Evolution into Space42

- Yahsat merged with Bayanat in Apr. 2024 to form Space42, combining satellite communications with geospatial AI capabilities.

Future Developments — What Comes Next

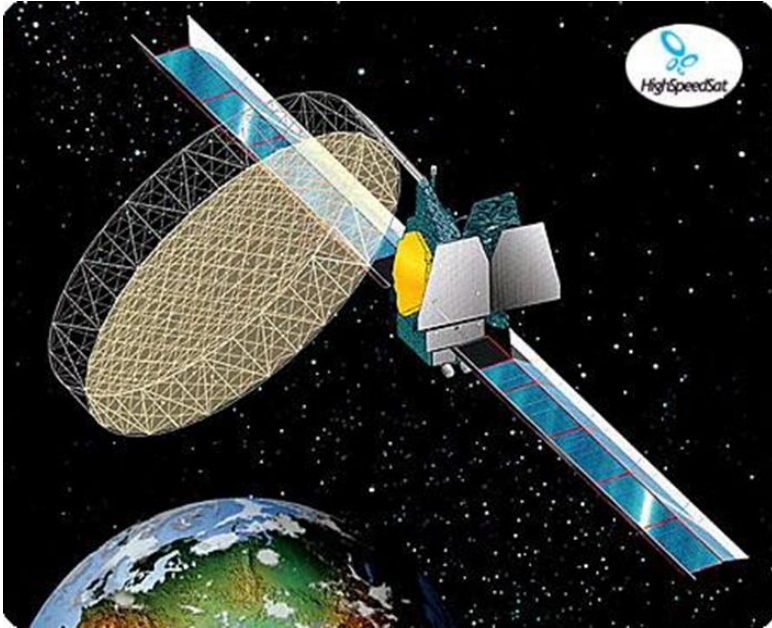


Fig. Thuraya-2: GEO mobile satellite system

Continuous GEO Coverage

- Stationary GEO operation avoids inter-satellite handoff and provides stable regional service.

Direct Handheld Connectivity

- L-band links make practical communication possible with low-power 2 W satellite phones.

Efficient Payload Design

- AstroMesh reflector + phased-array feed + spot beams concentrate capacity where demand is highest.

Overall, Thuraya-2 remains a strong example of how a single GEO satellite can deliver reliable mobile service in regions where terrestrial infrastructure is unavailable or impractical.